

FPT = PTIME for Homomorphism Problems on Sparse-Incidence and Bounded-Independence Patterns

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Abstract

Assuming the Exponential Time Hypothesis (ETH), fixed-parameter tractability and polynomial-time solvability coincide for homomorphism problems specified by classes of pattern hypergraphs of bounded incidence degeneracy or bounded primal independence number. In both cases, tractability is characterised by bounded fractional hypertree width. Grohe (JACM 2007) established the corresponding FPT–PTIME equivalence under bounded arity. Our result allows unbounded arity and covers important cases such as bounded-degree patterns and patterns whose incidence graphs exclude a fixed minor. Building on the recent fractional balanced-separator framework and rounding theorem of Korchemna et al. (FOCS 2024), we prove a near-linear bound on fractional hypertree width (fhw) in terms of adaptive width (adw). For every hypergraph H with $\text{adw}(H) \geq 2$,

$$\text{fhw}(H) = O(\lambda(H) \text{adw}(H) \log \text{adw}(H)),$$

where $\lambda(H) = \min\{\mu(H), \max\{1, \log \alpha(H)\}\}$, with $\mu(H)$ denoting incidence degeneracy and $\alpha(H)$ the independence number of the primal graph. As a further consequence, we obtain a corresponding FPT–PTIME collapse for exact homomorphism counting on every bounded- λ class. More generally, for every recursively enumerable class of pattern hypergraphs, fixed-parameter tractability of the parameterised homomorphism problem implies quasipolynomial-time solvability of the corresponding unparameterised problem, assuming ETH.

1 Introduction

We study the homomorphism problem between relational structures. For relational structures A, B over the same vocabulary, a homomorphism from A to B is a map from the domain of A to the domain of B that preserves every relation. We write $A \rightarrow B$ when such a map exists. This is known to generalise a wide range of common algorithmic problems. For instance, on graph structures, if B is fixed to be the k -clique K_k , then deciding $A \rightarrow K_k$ is equivalent to k -colourability of A . On the other hand, fixing $A = K_k$, we have that $K_k \rightarrow B$ is equivalent to the k -clique problem, and $K_k \rightarrow \bar{B}$ is the k -independent set problem.

In general, the problem is NP-complete. As the examples already suggest, it is of major interest to understand under which restrictions on the left-hand side (the *patterns*), or on the right-hand side (the *hosts* or *templates*), the problem is tractable. To simplify further discussion, we refer to the setting in which the templates belong to a fixed class as the constraint satisfaction problem (CSP). The complexity of restricting either side has been studied extensively, and for CSP this has culminated in a dichotomy theorem that precisely characterises for which classes of templates there are tractable algorithms, and for which the problem is NP-hard [5, 36]. Restricting the left-hand side is an entirely different problem. For instance, Bodirsky and Grohe [1] showed that no analogous dichotomy exists for arbitrary pattern classes. In this paper we study restrictions to the left-hand sides. Formally, for a class of hypergraphs $\mathcal{H}$ and H_A denoting the hypergraph structure of A :

$\text{Hom}(\mathcal{H})$

Input Finite relational structures A and B with $H_A \in \mathcal{H}$.

Question Does $A \rightarrow B$?

We write $p\text{-Hom}(\mathcal{H})$ for the parameterisation of $\text{Hom}(\mathcal{H})$ by the pattern A . In the database literature and in logic, $\text{Hom}(\mathcal{H})$ corresponds to the Boolean conjunctive query problem, with A being the query and B the database on which it is evaluated [6].

While there is no PTIME vs. NP-complete dichotomy for $\text{Hom}(\mathcal{H})$, there are still results on the limits of tractability. In a landmark result Grohe [14] proved that, for every recursively enumerable class of bounded-arity structures, the problem is fixed-parameter tractable exactly when it is feasible in polynomial time, assuming $\text{FPT} \neq \text{W}[1]$. Grohe’s theorem leaves open the case of unbounded arity, where treewidth no longer captures tractability. A structure consisting of a single tuple with r distinct entries has treewidth $r - 1$, although its homomorphism problem is easy.

This motivated more “hypergraph-native” variants of treewidth, beginning with hypertree width and generalised hypertree width (ghw) [12], and then fractional hypertree width (fhw) [15]. These notions are still based on tree decompositions, but introduce more complex cost functions for bags (in contrast to cardinality of the bag for treewidth). In particular, fractional hypertree width at most k means that there is a tree decomposition such that every bag has fractional edge-cover number at most k (formal definitions follow in Section 2). Analogously, generalised hypertree width (ghw) uses integral edge covers instead. Bounded fractional hypertree width remains the most general known sufficient condition on the pattern side for $\text{Hom}(\mathcal{H}) \in \text{PTIME}$ [15].

The tractability boundary extends further in the parameterised setting. Marx introduced *adaptive width* (adw) and *submodular width* (subw) [29, 30]. Treewidth and the hypertree widths have a single decomposition as a witness, with a small cardinality or cover in every bag. Adaptive and submodular width instead choose a decomposition separately for each cost function. This additional layer of quantification over sets of functions makes connecting the two kinds of width notions technically challenging. Remarkably, assuming ETH, $p\text{-Hom}(\mathcal{H})$ is in FPT exactly when $\mathcal{H}$ has bounded submodular width, or equivalently bounded adaptive width [30].

The standard inequalities are [15, 29, 30]

$$\text{adw}(H) \leq \text{subw}(H) \leq \text{fhw}(H) \leq \text{ghw}(H).$$

The reverse direction is much less understood. The additional quantification over functions, and the possibility of a different tree decomposition witness for each function, make adaptive and submodular width difficult to technically connect to fhw. Moreover, there are classes of bounded adaptive width and unbounded fractional hypertree width [29], so a reverse bound requires additional structure.

Few such comparisons are known. For hypergraphs of maximum vertex degree two, where each element occurs in at most two tuples, Lanzinger [25] showed under ETH that bounded fractional hypertree width and bounded submodular width coincide for structures with degree 2, i.e., every domain element occurs in at most 2 tuples. This conclusion follows through the complexity characterisations and gives no explicit quantitative bound between the widths. More recently, Lanzinger [26] gives quantitative comparisons in the form $\text{ghw}(H) = O(\text{subw}(H) \log \text{subw}(H))$ under two newly introduced structural conditions *bounded edge excess* and *private intersections*. For a detailed discussion of the connections to our main results, see Section 6. Furthermore, even small values of submodular width on graphs have required substantial analysis. A recent example is the work of Bringmann and Gorbachev [3], which classifies graph patterns of subquadratic complexity and relates their optimal exponents to submodular width.

Our main theorem gives a bound that closely couples fhw and adw, connecting these two different branches of hypergraph width measures. The two are linked via a factor $\lambda(H)$ that is the minimum

of the incidence degeneracy and the truncated logarithm of the independence number of the primal graph.

Theorem 1.1. *There is a universal constant $C > 0$ such that every finite hypergraph H with $\text{adw}(H) \geq 2$ satisfies*

$$\text{fhw}(H) \leq C\lambda(H)\text{adw}(H) \log \text{adw}(H).$$

When $\text{adw}(H) < 2$, $\text{fhw}(H) \leq C\lambda(H)$.

On a technical level we build on recent work of Korchemna et al. [23]. They introduce a framework of fractional balanced separators and an accompanying rounding theorem to convert fractional separators into integral separators in order to compute approximations of fhw efficiently. We instead observe that the space of fractional balanced separators is compact and convex. Through a minimax argument we then bound fractional balanced separators in terms of adaptive width. From this we are eventually able to show constructively that a tree decomposition adhering to the bound in Theorem 1.1 always exists.

Theorem 1.1 supplies the missing quantitative link between two well-developed theories. On classes of bounded λ , fractional hypertree width is within a logarithmic factor of adaptive width. For arbitrary hypergraphs, the bound $\lambda(H) \leq O(\log |V(H)|)$ still gives a polylogarithmic comparison when adaptive width is bounded.

This link allows us to combine classical tractability results based on fractional hypertree width with parameterised complexity results characterised by adaptive and submodular width. It also transfers bounds across the intermediate polymatroid width measures. As a result, we obtain the following consequences by combining results from the substantial bodies of work on the two sides of this comparison.

1) FPT = PTIME for bounded- λ hypergraph classes Together with the width hierarchy, the theorem shows that bounded fractional hypertree width, bounded submodular width, and bounded adaptive width are equivalent up to a $\log \text{adw}$ factor on every class with bounded λ . Since bounded subw is equivalent to the existence of an FPT algorithm and bounded fhw implies a polynomial-time algorithm, the two boundaries collapse.

Theorem 1.2. *Let $\mathcal{H}$ be a recursively enumerable hypergraph class of bounded λ . Assuming ETH,*

$$p\text{-Hom}(\mathcal{H}) \in \text{FPT} \iff \text{Hom}(\mathcal{H}) \in \text{PTIME}.$$

These conditions hold exactly when $\mathcal{H}$ has bounded fractional hypertree width.

We discuss properties studied in the literature that ensure bounded λ in the technical sections. For now, note that these include the widely studied case of bounded degree as well as all classes with sparse incidence graphs, i.e., bounded incidence degeneracy.

2) FPT implies quasipolynomial time Since $\lambda(H) \leq \log |V(H)|$ for $|V(H)| \geq 2$, the main comparison turns a constant adaptive-width bound into an $O(\log |V(H)|)$ fractional hypertree-width bound. This gives the following consequence. Write QP for quasipolynomial time.

Theorem 1.3. *Let $\mathcal{H}$ be a recursively enumerable hypergraph class. Assuming ETH,*

$$p\text{-Hom}(\mathcal{H}) \in \text{FPT} \Rightarrow \text{Hom}(\mathcal{H}) \in \text{QP}.$$

3) Exact counting on bounded- λ classes It is open in general whether fixed-parameter tractability of deciding $A \rightarrow B$ implies fixed-parameter tractability of counting all homomorphisms from A to B . We settle this question for bounded- λ hypergraph classes under ETH and obtain the stronger conclusion that both are equivalent to polynomial-time homomorphism counting. Write $\#\text{Hom}(\mathcal{H})$ for the problem of counting all homomorphisms from patterns in $\mathcal{H}$, and write $p\text{-}\#\text{Hom}(\mathcal{H})$ for its parameterisation by the pattern. For every recursively enumerable class $\mathcal{H}$ of bounded λ , assuming ETH,

$$p\text{-Hom}(\mathcal{H}) \in \text{FPT} \iff p\text{-}\#\text{Hom}(\mathcal{H}) \in \text{FPT} \iff \#\text{Hom}(\mathcal{H}) \in \text{PTIME}.$$

Additionally, our main result addresses open questions regarding the relationship between the various intermediate widths defined from normal, entropic, almost-entropic, and sharp-submodular polymatroids. These quantities arise in database theory in the analysis of query algorithms, semiring aggregation, and degree-aware worst-case join-size bounds [19, 21, 35]. On bounded- λ classes, they all lie within a logarithmic factor of the width over the modular polymatroid (the adaptive width).

Organisation of the paper. Section 2 introduces technical preliminaries, the parameter λ , and the width measures used throughout the paper. Section 3 proves Theorem 1.1. Section 4 derives the complexity-theoretic consequences for hypergraph classes and extends them to classes of structures. Section 5 develops the consequences for exact homomorphism counting and the polymatroid width hierarchy. Section 6 discusses the relationship of our results to closely related works in the literature. Section 7 concludes with open problems.

2 Preliminaries

All logarithms are to base two unless otherwise specified.

2.1 Relational Structures and Hypergraphs

A hypergraph is a pair $H = (V(H), E(H))$, with $E(H) \subseteq 2^{V(H)}$. We refer to $V(H)$ as the vertices of H and $E(H)$ as the edges of H . All hypergraphs are finite, have no empty edges, and have every vertex contained in an edge. We assume input hypergraphs are nonempty and connected, but allow induced subhypergraphs to be empty or disconnected. The *rank* $\text{rank}(H)$ is the maximum hyperedge size. The *(vertex) degree* $\Delta(H)$ is the largest number of hyperedges containing one vertex.

For $W \subseteq V(H)$, the *induced subhypergraph* $H[W]$ has vertex set W and hyperedge set

$$E(H[W]) = \{e \cap W \mid e \in E(H), e \cap W \neq \emptyset\}.$$

Note that this differs from other common subhypergraph notions where partial edges are not retained. In the context of relational structures, this corresponds to projecting the structure to some subset of the domain elements.

A finite relational signature τ assigns an arity r_R to each relation symbol R . A finite τ -structure A consists of a finite domain $V(A)$ and a relation $R^A \subseteq V(A)^{r_R}$ for every $R \in \tau$. Relations are represented by explicit lists of tuples, and $\|A\|$ denotes the encoding length. A homomorphism $h : A \rightarrow B$ is a map $V(A) \rightarrow V(B)$ such that

$$(a_1, \dots, a_{r_R}) \in R^A \implies (h(a_1), \dots, h(a_{r_R})) \in R^B$$

for every $R \in \tau$.

A relational structure naturally induces a hypergraph that captures the connections between its elements. The hypergraph H_A has one vertex for each domain element occurring in a tuple. For every tuple $\mathbf{a} \in R^A$, it has the hyperedge consisting of the entries of $\mathbf{a}$. Different tuples that induce the same set contribute one hyperedge.

The $\lambda(H)$ parameter We move on to defining the parameter $\lambda(H)$ in [Theorem 1.1](#). It is the minimum of two independent hypergraph invariants and originates in a technical result in [\[23\]](#).

The *primal graph* $P(H)$ has vertex set $V(H)$ and joins two distinct vertices when they occur together in a hyperedge. Write $\alpha(H)$ for the independence number of $P(H)$, i.e., the maximum number of vertices such that no two vertices are in the same hyperedge. The *incidence graph* $I(H)$ is the bipartite graph with parts $V(H)$ and $E(H)$ and incidence as adjacency. Define

$$\mu(H) = \max_{\substack{F \subseteq I(H) \\ V(F) \neq \emptyset}} \min_{x \in V(F)} \deg_F(x).$$

Thus $\mu(H)$ is the least integer d such that every nonempty subgraph of $I(H)$ has a vertex of degree at most d , called the *incidence degeneracy* of H [\[31\]](#). Then define

$$\lambda(H) = \min\{\mu(H), \max\{1, \log \alpha(H)\}\}. \quad (1)$$

Note that both μ and α are nonincreasing under taking induced subhypergraphs.

2.2 Tree Decompositions and Width Functions

A tree decomposition of a hypergraph H is a pair (T, B) , where T is a tree and $B : V(T) \rightarrow 2^{V(H)}$ such that:

1. for each $e \in E(H)$ there is a node $u \in V(T)$ such that $e \subseteq B(u)$.
2. for every vertex $v \in V(H)$, the set of nodes $\{u \in V(T) \mid v \in B(u)\}$ is nonempty and forms a connected subgraph of T .

Write $\mathcal{T}(H)$ for the set of tree decompositions of H and $\text{width}_b(T, B) = \max_{t \in V(T)} b(B_t)$ for the width of a decomposition under a function $b : 2^{V(H)} \rightarrow \mathbb{R}_{\geq 0}$. For a nonempty family $\mathcal{F}$ of such functions, its $\mathcal{F}$ -width is

$$\text{width}_{\mathcal{F}}(H) := \sup_{b \in \mathcal{F}} \min_{(T, B) \in \mathcal{T}(H)} \max_{t \in V(T)} b(B_t). \quad (2)$$

For $X \subseteq V(H)$, the *fractional edge-cover number* $\rho_H^*(X)$ is the minimum of $\sum_e y_e$ over $y_e \geq 0$ satisfying $\sum_{e \ni v} y_e \geq 1$ for every $v \in X$. Requiring $y_e \in \{0, 1\}$ gives the integral edge-cover number $\rho_H(X)$. Fractional and generalised hypertree width are

$$\text{fhw}(H) = \text{width}_{\{\rho_H^*\}}(H), \quad \text{ghw}(H) = \text{width}_{\{\rho_H\}}(H).$$

Treewidth is $\text{width}_{\{X \mapsto |X|\}}(H) - 1$. A fractional hypertree decomposition (FHD) supplies each bag with a fractional edge cover. We use optimum bag covers and write $\text{width}(\mathcal{D}) = \text{width}_{\rho_H^*}(T, B)$ for $\mathcal{D} = (T, B)$.

A *polymatroid* is a function $b : 2^{V(H)} \rightarrow \mathbb{R}_{\geq 0}$ with $b(\emptyset) = 0$ that is monotone and submodular. That is,

$$b(X) \leq b(Y) \quad \text{whenever } X \subseteq Y, \quad \text{and} \quad b(X \cup Y) + b(X \cap Y) \leq b(X) + b(Y)$$

for all $X, Y \subseteq V(H)$. It is *edge-dominated* if $b(e) \leq 1$ for every $e \in E(H)$. Let $\mathcal{S}_H$ be the family of edge-dominated polymatroids and let $\mathcal{M}_H \subseteq \mathcal{S}_H$ consist of the modular functions $b(X) = \sum_{v \in X} w_v$, where $w_v \geq 0$ and $\sum_{v \in e} w_v \leq 1$ for every hyperedge e . We call such a vector w an *edge-dominated modular weighting*. Adaptive width and submodular width are defined by [29, 30]

$$\text{adw}(H) = \text{width}_{\mathcal{M}_H}(H), \quad \text{subw}(H) = \text{width}_{\mathcal{S}_H}(H).$$

It is well known that $\text{adw}(H) \leq \text{subw}(H) \leq \text{fhw}(H) \leq \text{ghw}(H)$ [15, 29, 30]. For $X \subseteq W \subseteq V(H)$, fractional cover cost satisfies

$$\rho_{H[W]}^*(X) = \rho_H^*(X). \quad (3)$$

Indeed, a fractional cover of H induces one of $H[W]$ by aggregating the weights of all hyperedges with the same nonempty intersection with W and discarding hyperedges disjoint from W . Conversely, a fractional cover of $H[W]$ lifts to H by choosing, for every $f \in E(H[W])$, one hyperedge $e_f \in E(H)$ with $e_f \cap W = f$ and assigning the weight of f to e_f . The first operation does not increase total weight, and the second preserves it, which proves (3). Moreover, it is straightforward that adaptive width is nonincreasing under taking induced subhypergraphs, since modular weightings extend by zero and tree decompositions restrict to vertex subsets.

3 Fractional Separators and Adaptive Width

Abo Khamis et al. [22] observed that fractional edge-cover duality lets us naturally view adaptive width and fractional hypertree width as opposite orders of optimisation.

$$\text{adw}(H) = \max_{b \in \mathcal{M}_H} \min_{(T, B) \in \mathcal{T}(H)} \text{width}_b(T, B), \quad \text{fhw}(H) = \min_{(T, B) \in \mathcal{T}(H)} \max_{b \in \mathcal{M}_H} \text{width}_b(T, B).$$

Here $\mathcal{M}_H$ is the family of edge-dominated modular functions. Weak minimax recovers only the familiar inequality $\text{adw}(H) \leq \text{fhw}(H)$. Strong minimax does not apply directly. The space of tree decompositions has no useful convex structure, and $\text{width}_b(T, B) = \max_{t \in V(T)} b(B_t)$ is the maximum of linear functions of b , hence convex rather than concave in the maximising variable.

The difference between the two optimisation orders is that adaptive width allows us to choose a different decomposition for each b , whereas fractional hypertree width asks for one decomposition whose bags are cheap for every b . Our proof bridges this difference through balanced separators. They are simpler objects than decompositions, but retain the two properties we need. A decomposition supplies a balanced separator, and suitable balanced separators can be assembled recursively into a decomposition. We will do so through a relaxed version of balanced separators that provides us with the appropriate properties for a minimax argument connecting to adw .

In Section 3.1, we recall fractional balanced separators and their fractional edge-cover cost. In Section 3.2, we use minimax to show that adaptive width bounds the cost of such fractional balanced separators in every induced subhypergraph. In Section 3.3, we round these separators and assemble them recursively into a tree decomposition, proving Theorem 1.1.

3.1 Fractional Balanced Separators

We first recall the hypergraph fractional balanced-separator relaxation of Korchemna et al. [23, Definition 5.2]. It is a hyperedge-weighted analogue of the spreading-metric formulation of ρ -separators due to Even, Naor, Rao, and Schieber [11], building on the metric and region-growing framework of Leighton and Rao [27].

In the following, we consider functions $x : V(J) \rightarrow [0, 1]$ of vertex demands that we interpret as *fractional separators* on a hypergraph J . This relaxes the classical notion of a separator, in which vertices either belong to the separator or do not. For $S \subseteq V(J)$, we write $\mathbf{1}_S$ for the vertex-demand function that maps vertices in S to 1 and all other vertices to 0. For vertex demands x , define

$$\rho_J^*(x) := \min \left\{ \sum_{e \in E(J)} y(e) : y : E(J) \rightarrow \mathbb{R}_{\geq 0}, \quad \sum_{e \ni v} y(e) \geq x(v) \quad \forall v \in V(J) \right\}.$$

Thus a fractional edge cover need only satisfy the vertex demand $x(v)$, rather than $\sum_{e \ni v} y(e) \geq 1$ uniformly. When x is a fractional separator, this will eventually correspond to the “cost” of the separator.

For vertices u, v , let $\text{paths}(P(J); u, v)$ be the set of simple paths in $P(J)$ with endpoints u and v . We are interested in the minimum-demand path between any pair of vertices. To that end, define

$$\text{dist}_{J,x}(u, v) := \min_{Q \in \text{paths}(P(J); u, v)} \sum_{z \in V(Q)} x(z), \quad \text{dist}_{J,x}^*(u, v) := \min\{1, \text{dist}_{J,x}(u, v)\}.$$

Note that the sum includes both endpoints and we consider the minimum of the empty set to be ∞ . When x is seen as a fractional separator, the distance between u and v is a measure of how separated they are, with values above 1 meaning they are fully separated. We therefore use the auxiliary quantity $\text{dist}_{J,x}^*$, which is the truncation of $\text{dist}_{J,x}$ to $[0, 1]$. Note that if $x = \mathbf{1}_S$ for some $S \subseteq V(J)$, then $\text{dist}_{J,x}^*(u, v) = 0$ exactly when u and v are connected in $P(J) - S$, and it equals 1 otherwise. We extend this distance to hyperedges naturally. For hyperedges e, f , define $\text{dist}_{J,x}^*(e, f) := \min_{u \in e, v \in f} \text{dist}_{J,x}^*(u, v)$.

We also consider a weighting $\gamma : E(J) \rightarrow [0, 1]$ that records the contribution of each edge to the balance condition. In particular, let $\gamma : E(J) \rightarrow [0, 1]$ be nonzero and write $\Gamma = \sum_e \gamma(e)$. For $\theta \in (0, 1)$, a function $x : V(J) \rightarrow [0, 1]$ is a *fractional (γ, θ) -balanced separator* if

$$\sum_{f \in E(J)} \gamma(f) \text{dist}_{J,x}^*(e, f) \geq (1 - \theta)\Gamma \quad \forall e \in E(J). \quad (4)$$

Intuitively, the left-hand side can be read as a measure of how strongly edge e is separated from all other edges f (weighted by γ) by the fractional separator x .

When the separator is not fractional, i.e., it is a set $S \subseteq V(J)$, we say that S is *(γ, θ) -balanced* if every component C of $J - S$ satisfies

$$\sum_{\substack{e \in E(J) \\ e \cap C \neq \emptyset}} \gamma(e) \leq \theta\Gamma. \quad (5)$$

Lemma 3.1 ([23, Fact 5.4]). *Let J be a hypergraph, $S \subseteq V(J)$, $\gamma : E(J) \rightarrow [0, 1]$ nonzero, and $\theta \in (0, 1)$. Then $\mathbf{1}_S$ is a fractional (γ, θ) -balanced separator if and only if S is (γ, θ) -balanced.*

So far, we have followed the definitions of [23]. The linear-programming dual form of ρ^* will also be important for our result. It is

$$\rho_J^*(x) = \max \left\{ \sum_{v \in V(J)} w_v x(v) : w_v \geq 0, \quad \sum_{v \in e} w_v \leq 1 \quad \forall e \in E(J) \right\}. \quad (6)$$

Moreover, we will be interested in the least ρ^* necessary for a (γ, θ) -balanced fractional separator, for fixed γ, θ . Define the optimum fractional cost of one separator instance by

$$\beta_{J,\theta}(\gamma) := \min \{ \rho_J^*(x) \mid x \text{ is a fractional } (\gamma, \theta)\text{-balanced separator} \}.$$

Finally, we use fractional balanced separators as an intermediate object between adaptive width and fractional hypertree width. The recursive construction of a tree decomposition requires such separators in every induced subhypergraph and for every weighting of its hyperedges. For this process, we define

$$\text{fsep}_\theta(H) := \sup_{\substack{W \subseteq V(H) \\ 0 \neq \gamma: E(H[W]) \rightarrow [0,1]}} \beta_{H[W], \theta}(\gamma).$$

Equivalently, $\text{fsep}_\theta(H)$ is the least b such that every induced subhypergraph J of H and every nonzero hyperedge weighting γ admit a fractional (γ, θ) -balanced separator x with $\rho_J^*(x) \leq b$.

3.2 Adaptive Width and Fractional Separators

The convexity of fractional balanced separators allows minimax to compare their optimum cost with adaptive width. We prove $\text{fsep}_{1/2}(H) \leq \text{adw}(H)$.

Lemma 3.2. *Let J be a hypergraph, let $\gamma: E(J) \rightarrow [0, 1]$ be nonzero, and let $\theta \in (0, 1)$. The set of fractional (γ, θ) -balanced separators is nonempty, compact, and convex.*

Proof. For $e, f \in E(J)$, let $\mathcal{P}(e, f)$ be the set of simple paths in $P(J)$ joining a vertex of e to a vertex of f , including paths of length zero. By definition,

$$\text{dist}_{J,x}^*(e, f) = \min \left(\{1\} \cup \left\{ \sum_{v \in V(Q)} x(v) \mid Q \in \mathcal{P}(e, f) \right\} \right).$$

Thus $x \mapsto \text{dist}_{J,x}^*(e, f)$ is continuous and concave, since it is the minimum of finitely many affine functions. Since $\gamma \geq 0$, the left-hand side of each balance constraint (4) is also continuous and concave. Each constraint therefore defines a closed convex subset of $[0, 1]^{V(J)}$. Their intersection is convex and is a closed subset of the compact cube $[0, 1]^{V(J)}$, hence compact. It is nonempty because it contains the all-one vector. $\square$

Lemma 3.3. *Let J be a hypergraph, let $\gamma: E(J) \rightarrow [0, 1]$ be nonzero, and let $\theta \in (0, 1)$. There is $b \in \mathcal{M}_J$ such that*

$$\sum_{v \in V(J)} b(\{v\})x(v) \geq \beta_{J,\theta}(\gamma)$$

for every fractional (γ, θ) -balanced separator x .

Proof. Let $\mathcal{X}$ be the set of fractional (γ, θ) -balanced separators. Identify each $b \in \mathcal{M}_J$ with its singleton values $(b(\{v\}))_{v \in V(J)}$. By Equation (6),

$$\beta_{J,\theta}(\gamma) = \min_{x \in \mathcal{X}} \max_{b \in \mathcal{M}_J} \sum_v b(\{v\})x(v).$$

By Lemma 3.2, $\mathcal{X}$ is nonempty, compact, and convex. In these coordinates, $\mathcal{M}_J$ is a closed convex polytope containing the zero vector. Every vertex lies in a hyperedge, so $\mathcal{M}_J \subseteq [0, 1]^{V(J)}$ and $\mathcal{M}_J$ is compact. Furthermore, the map $(x, b) \mapsto \sum_v b(\{v\})x(v)$ is continuous and bilinear. A standard application of the minimax theorem over convex sets [34] thus gives

$$\beta_{J,\theta}(\gamma) = \max_{b \in \mathcal{M}_J} \min_{x \in \mathcal{X}} \sum_v b(\{v\})x(v). \quad (7)$$

Any maximising b has the required property. $\square$

Lemma 3.4. *For every hypergraph H , $\text{fsep}_{1/2}(H) \leq \text{adw}(H)$.*

Proof. Fix $W \subseteq V(H)$, let $J = H[W]$, and fix a nonzero weighting $\gamma : E(J) \rightarrow [0, 1]$. Write $\Gamma = \sum_{e \in E(J)} \gamma(e)$ and $\beta = \beta_{J, 1/2}(\gamma)$. Choose $b \in \mathcal{M}_J$ from Lemma 3.3, so that $\sum_{v \in V(J)} b(\{v\})x(v) \geq \beta$ for every fractional $(\gamma, 1/2)$ -balanced separator x .

Let (T, B) be any tree decomposition of J . For each $e \in E(J)$, choose one node $a(e) \in V(T)$ with $e \subseteq B_{a(e)}$. Give each node s weight $q(s) = \sum_{e \in E(J), a(e)=s} \gamma(e)$, adding the weights of all hyperedges assigned to s . Thus $\sum_{s \in V(T)} q(s) = \Gamma$.

We first show that there is a node t such that every component K of $T - t$ satisfies

$$\sum_{s \in V(K)} q(s) \leq \Gamma/2.$$

To construct t , start at any node s . Whenever a component of $T - s$ has total q -weight greater than $\Gamma/2$, move to the neighbour of s in that component. After a move from s to s' , the component of $T - s'$ containing s has weight less than $\Gamma/2$, so the next move cannot return to s . A walk in a tree that never immediately reverses an edge cannot revisit a node. Since T is finite, the process terminates at a node t with the required property.

We next show that B_t is $(\gamma, 1/2)$ -balanced. Let C be the vertex set of a component of $P(J) - B_t$. For each $v \in C$, the set $T_v = \{s \in V(T) \mid v \in B_s\}$ is connected and avoids t , so it lies in one component of $T - t$. If $u, v \in C$ are adjacent in $P(J)$, some hyperedge contains both, and a bag containing that hyperedge belongs to $T_u \cap T_v$. Since C is connected, all sets T_v with $v \in C$ lie in a single component K of $T - t$. For every hyperedge e meeting C , choose $v \in e \cap C$. Then $a(e) \in T_v \subseteq V(K)$, since $e \subseteq B_{a(e)}$. Therefore

$$\sum_{e \in E(J), e \cap C \neq \emptyset} \gamma(e) \leq \sum_{s \in V(K)} q(s) \leq \Gamma/2$$

as required by (5).

By Lemma 3.1, $\mathbf{1}_{B_t}$ is a fractional $(\gamma, 1/2)$ -balanced separator. The choice of b gives $b(B_t) = \sum_{v \in V(J)} b(\{v\})\mathbf{1}_{B_t}(v) \geq \beta$. Since (T, B) was arbitrary and $b \in \mathcal{M}_J$, we obtain

$$\beta \leq \min_{(T, B) \in \mathcal{T}(J)} \max_{s \in V(T)} b(B_s) \leq \text{adw}(J) \leq \text{adw}(H).$$

Taking the supremum over W and γ gives $\text{fsep}_{1/2}(H) \leq \text{adw}(H)$. □

3.3 Rounding Fractional Separators and Decompositions

Korchemna et al. [23] round fractional half-balanced separators to integral separators with controlled fractional edge-cover number. Combined with $\text{fsep}_{1/2}(H) \leq \text{adw}(H)$, their result supplies the separators needed for the recursive decomposition.

Proposition 3.5 ([23, Theorem 5.6]). *Let J be a hypergraph, let $0 \neq \gamma : E(J) \rightarrow [0, 1]$, and let x be a fractional $(\gamma, 1/2)$ -balanced separator. There is an integral $(\gamma, 5/6)$ -balanced separator $S \subseteq V(J)$ satisfying*

$$\rho_J^*(S) \leq C_0 \lambda(J) \rho_J^*(x) \log \max\{2, \rho_J^*(x)\}$$

for a universal constant C_0 .

Proof. If $\emptyset$ is $(\gamma, 5/6)$ -balanced, take $S = \emptyset$. Otherwise set $r = \rho_J^*(x)$. If $r = 0$, then $x = 0$, so $\mathbf{1}_\emptyset = x$ is fractionally $(\gamma, 1/2)$ -balanced. By Lemma 3.1, $\emptyset$ is integrally $(\gamma, 1/2)$ -balanced and hence $(\gamma, 5/6)$ -balanced, a contradiction. Thus $r > 0$. Write $\text{supp } x = \{v \in V(J) \mid x(v) > 0\}$. The cited theorem with $\varphi = 1/2$ produces a $(\gamma, 5/6)$ -balanced separator and gives

$$\rho_J^*(S) \leq (\min\{8 + 4 \ln \alpha(J[\text{supp } x]), 6\mu(J)\} + 1)(88 + 16 \log(2r))r.$$

Put $A = \alpha(J[\text{supp } x])$. Then $A \leq \alpha(J)$. Since $\mu(J) \geq 1$, the first factor is at most $7\mu(J)$ and at most $9 + 4 \ln \alpha(J)$. It is therefore $O(\lambda(J))$. Moreover,

$$88 + 16 \log(2r) \leq 88 + 16 \max\{0, \log(2r)\} = O(\log \max\{2, r\}).$$

Absorbing the constants into C_0 proves the claim. $\square$

The parameter λ enters only through this rounding step. Any alternative rounding bound would propagate through the same recursion.

We now turn these rounded separators into a tree decomposition using the recursive framework of Robertson and Seymour [33]. A related separator lemma for monotone subadditive set functions appears in Marx [29, Section 5.2].

Lemma 3.6. *Let H be a hypergraph, let $\eta \in (0, 1)$, and let $r \geq 0$. Suppose that for every $W \subseteq V(H)$ and every nonzero $\gamma : E(H[W]) \rightarrow [0, 1]$, there is a (γ, η) -balanced separator $S \subseteq W$ with $\rho_{H[W]}^*(S) \leq r$. Then*

$$\text{fhw}(H) \leq \frac{2(2 - \eta)}{1 - \eta} r.$$

Proof. Set $\Lambda = 2r/(1 - \eta)$. We prove by induction on $|W \setminus Z|$ that whenever $Z \subseteq W \subseteq V(H)$ and $\rho_H^*(Z) \leq \Lambda$, the hypergraph $H[W]$ has a tree decomposition with a bag containing Z and every bag B_t satisfying $\rho_H^*(B_t) \leq \Lambda + 2r$. The set Z is the boundary through which this decomposition will attach to its parent. We use a separator S to control the cover cost of each child boundary and a separator R to ensure that each child has fewer vertices outside its boundary.

If $W = Z$, take the one-bag decomposition with bag Z . Its bag cost is at most Λ , so the claim holds. Suppose that $W \neq Z$ and let $J = H[W]$. If $Z = \emptyset$, set $S = \emptyset$. If $Z \neq \emptyset$, choose an optimum fractional edge cover $\gamma : E(J) \rightarrow [0, 1]$ of Z . Because $Z \subseteq V(J)$ and fractional cover cost is invariant under taking induced subhypergraphs,

$$\sum_{e \in E(J)} \gamma(e) = \rho_J^*(Z) = \rho_H^*(Z).$$

Since $Z \neq \emptyset$, the weighting γ is nonzero, and by the assumption in the statement there is a (γ, η) -balanced separator $S \subseteq W$ with $\rho_J^*(S) \leq r$ and thus also $\rho_H^*(S) \leq r$.

For each $v \in W \setminus Z$, choose $e_v \in E(J)$ with $v \in e_v$, and define $\pi : E(J) \rightarrow [0, 1]$ by

$$\pi(e) = \frac{|\{v \in W \setminus Z : e_v = e\}|}{|W \setminus Z|}.$$

Since $\sum_e \pi(e) = 1$, the assumption in the statement gives a (π, η) -balanced separator $R \subseteq W$ with $\rho_J^*(R) \leq r$. Therefore also $\rho_H^*(R) \leq r$. Set $B = Z \cup S \cup R$. Subadditivity of ρ^* gives us

$$\rho_H^*(B) \leq \rho_H^*(Z) + \rho_H^*(S) + \rho_H^*(R) \leq \Lambda + 2r.$$

We move on to constructing our tree decomposition by separating J by B . For each component K of $J - B$, define its boundary and child vertex set as

$$Z_K := B \cap N_{P(J)}(K), \quad W_K := K \cup Z_K.$$

For our application of the induction hypothesis we will want to bound $\rho_H^*(Z_K)$ for each K by Λ . If $Z = \emptyset$, then $B = R$ and $Z_K \subseteq R$ and therefore directly $\rho_H^*(Z_K) \leq r \leq \Lambda$.

If instead $Z \neq \emptyset$, let C be the component of $J - S$ containing K . We have $K \subseteq C$, but equality need not hold because vertices of $Z \cup R$ can join several components of $J - B$ in $J - S$. Moreover, we have $Z_K \subseteq S \cup R \cup (Z \cap C)$. Indeed, if $z \in Z_K \setminus (S \cup R)$, then $z \in B \setminus (S \cup R) \subseteq Z$, and adjacency to $K \subseteq C$ in $J - S$ implies $z \in C$.

Restricting γ to edges meeting C still covers $Z \cap C$, since every edge containing a vertex of $Z \cap C$ meets C . The (γ, η) -balance of S therefore gives

$$\rho_H^*(Z \cap C) \leq \sum_{e \cap C \neq \emptyset} \gamma(e) \leq \eta \sum_{e \in E(J)} \gamma(e) = \eta \rho_H^*(Z) \leq \eta \Lambda.$$

Consequently,

$$\rho_H^*(Z_K) \leq \rho_H^*(S) + \rho_H^*(R) + \rho_H^*(Z \cap C) \leq 2r + \eta \Lambda = \Lambda,$$

where the last equality follows from $\Lambda = 2r/(1 - \eta)$.

Let D be the component of $J - R$ containing K . Every $v \in K$ belongs to $W \setminus Z$ and has $e_v \cap D \neq \emptyset$. Thus

$$\frac{|K|}{|W \setminus Z|} \leq \sum_{e \cap D \neq \emptyset} \pi(e) \leq \eta < 1.$$

Hence $|W_K \setminus Z_K| = |K| < |W \setminus Z|$. Since $\rho_H^*(Z_K) \leq \Lambda$, the induction hypothesis applied to (W_K, Z_K) gives a tree decomposition (T_K, B^K) of $H[W_K]$ containing an attachment node u_K whose bag includes Z_K . Every bag of this decomposition has fractional cover cost at most $\Lambda + 2r$.

Take disjoint copies of the trees T_K . Add a new node t with bag B and join t to the attachment node u_K of T_K . The resulting graph is a tree. Every vertex of $W \setminus B$ belongs to exactly one component K and hence to exactly one child instance. Every vertex of B occurs at t . If it also occurs in the child for K , then it belongs to $B \cap W_K = Z_K$ and hence occurs at the attachment node. It is straightforward to verify that this newly constructed decomposition satisfies the connectedness condition.

Let $e \in E(J)$. If $e \subseteq B$, then the new bag covers e . Otherwise, the vertices of $e \setminus B$ lie in one component K of $J - B$, since a hyperedge is a clique in $P(J)$. Every vertex of $e \cap B$ lies in Z_K , so $e \subseteq W_K$. Hence e is a hyperedge of $H[W_K]$ and is covered by the corresponding child decomposition. This proves the induction claim.

Apply the induction with $W = V(H)$ and $Z = \emptyset$. The resulting width is at most

$$\Lambda + 2r = \frac{2(2 - \eta)}{1 - \eta} r.$$

□

Theorem 1.1. *There is a universal constant $C > 0$ such that every finite hypergraph H with $\text{adw}(H) \geq 2$ satisfies*

$$\text{fhw}(H) \leq C \lambda(H) \text{adw}(H) \log \text{adw}(H).$$

When $\text{adw}(H) < 2$, $\text{fhw}(H) \leq C \lambda(H)$.

Proof of Theorem 1.1. Set $a = \max\{2, \text{adw}(H)\}$. Fix $W \subseteq V(H)$, put $J = H[W]$, and fix a nonzero weighting $\gamma : E(J) \rightarrow [0, 1]$. Let x be a fractional $(\gamma, 1/2)$ -balanced separator satisfying $\rho_J^*(x) = \beta_{J,1/2}(\gamma)$. Such an x exists by Lemma 3.2 and continuity of ρ_J^* . By the definition of $\text{fsep}_{1/2}$ and Lemma 3.4,

$$\rho_J^*(x) = \beta_{J,1/2}(\gamma) \leq \text{fsep}_{1/2}(H) \leq \text{adw}(H) \leq a.$$

Since $P(J) = P(H)[W]$, every independent set of $P(J)$ is independent in $P(H)$, and hence $\alpha(J) \leq \alpha(H)$. Moreover, $I(J)$ is isomorphic to a subgraph of $I(H)$. To see this, choose for each $f \in E(J)$ an edge $e_f \in E(H)$ with $f = e_f \cap W$. Distinct edges of J have distinct chosen preimages, and the map fixing the vertices in W and sending f to e_f preserves every incidence. Thus $\mu(J) \leq \mu(H)$. Together with the bound on $\alpha(J)$, this gives

$$\lambda(J) = \min\{\mu(J), \max\{1, \log \alpha(J)\}\} \leq \min\{\mu(H), \max\{1, \log \alpha(H)\}\} = \lambda(H).$$

As $a \geq 2$, Proposition 3.5 tells us that $\rho_J^*(S) \leq C_0 \lambda(H) a \log a$ for some $(\gamma, 5/6)$ -balanced separator S . The choices of W and γ were arbitrary, so Lemma 3.6 applies with $\eta = 5/6$ and $r = C_0 \lambda(H) a \log a$. Therefore

$$\text{fhw}(H) \leq 14r \leq C \lambda(H) a \log a.$$

For $\text{adw}(H) \geq 2$ this is the stated bound, and for $\text{adw}(H) < 2$ it gives $\text{fhw}(H) = O(\lambda(H))$. $\square$

Since $\alpha(H) \leq |V(H)|$, and hence $\lambda(H) \leq \log |V(H)|$, we also obtain the following surprising corollary.

Corollary 3.7. *For every n -vertex hypergraph H with $\text{adw}(H) \geq 2$,*

$$\text{fhw}(H) \leq C \text{adw}(H) \log \text{adw}(H) \log n$$

for a universal constant C .

4 Complexity Consequences

We now translate Theorem 1.1 into complexity statements for homomorphism problems. For hypergraph classes, fixed-parameter tractability is characterised by bounded submodular width, while the standard polynomial-time algorithms require bounded fractional hypertree width. These results do not combine directly, since $\text{subw} \leq \text{fhw}$ and the reverse inequality fails in general. Our width comparison supplies the missing reverse bound under control of λ . It yields an $\text{FPT} = \text{PTIME}$ collapse on bounded- λ classes and a quasipolynomial algorithm for every FPT hypergraph class. We then treat structure classes of bounded incidence degeneracy, where the relevant widths are those of the cores but computing the core itself is intractable.

4.1 Hypergraph Classes

First, recall that, given an FHD $\mathcal{D}$ of width k for H_A , deciding $A \rightarrow B$ takes time $\text{poly}(\|A\| + |\mathcal{D}| + \|B\|) \|B\|^{O(k)}$ [15]. Computing an optimum FHD is NP-hard even when the width is bounded by a fixed constant [13]. However, Marx's cubic approximation computes an FHD of width $O(k^3)$ in time $\|H_A\|^{O(k^3)}$ whenever $\text{fhw}(H_A) \leq k$ [28, Theorem 4.1]. At the cost of a cubic overhead in the exponent, one still obtains a polynomial-time algorithm for every fixed k .

On classes of bounded λ , Theorem 1.1 and the width hierarchy make bounded fractional hypertree width equivalent to bounded submodular width, which characterises fixed-parameter tractability under ETH [30]. Theorem 1.1 directly yields the $\text{FPT} = \text{PTIME}$ collapse in this setting.

Theorem 4.1. *Let $\mathcal{H}$ be a recursively enumerable hypergraph class with bounded λ . Assuming ETH, the following are equivalent.*

- (i) $\mathcal{H}$ has bounded fractional hypertree width.
- (ii) $\mathcal{H}$ has bounded submodular width.
- (iii) $p\text{-Hom}(\mathcal{H}) \in \text{FPT}$.
- (iv) $\text{Hom}(\mathcal{H}) \in \text{PTIME}$.

Proof. The width hierarchy $\text{adw}(H) \leq \text{subw}(H) \leq \text{fhw}(H)$ together with Theorem 1.1 gives (i) $\Leftrightarrow$ (ii). Marx [30] showed that (ii) $\Leftrightarrow$ (iii) under ETH. For (i) $\Rightarrow$ (iv), Grohe and Marx prove directly that $\text{Hom}(\mathcal{H}) \in \text{PTIME}$ [15, Corollary 4.11]. Every polynomial-time algorithm is fixed-parameter tractable, giving (iv) $\Rightarrow$ (iii). $\square$

This also proves Theorem 1.2.

Proof of Theorem 1.3. Assume ETH and $p\text{-Hom}(\mathcal{H}) \in \text{FPT}$. By [30], $\mathcal{H}$ has bounded submodular width, and hence bounded adaptive width. Let c be some constant bound on the adaptive width of $\mathcal{H}$.

On input (A, B) with $H_A \in \mathcal{H}$, let $N = \|A\| + \|B\|$. By Theorem 1.1 and $\lambda(H_A) = O(\log N)$, we have $\text{fhw}(H_A) \leq c' \log N$ where c' depends on c . The approximation algorithm from [28, Theorem 4.1] constructs an FHD of width $O(k^3)$ in time $\|H_A\|^{O(k^3)}$ whenever $\text{fhw}(H_A) \leq k$. Taking $k = c' \log N$ and evaluating (A, B) using the resulting decomposition gives total running time $N^{O(\log^3 N)}$. $\square$

4.2 Widths of Cores and Structure Classes

In the following, for a relational structure A and a hypergraph width function w , we write $w(A)$ for $w(H_A)$.

An endomorphism of A is a homomorphism from A to itself. A *core* of A is an endomorphic image of minimum domain cardinality. It is unique up to isomorphism, and we denote it by $\text{core}(A)$ (see also [6]). Since A and $\text{core}(A)$ are homomorphically equivalent,

$$A \rightarrow B \iff \text{core}(A) \rightarrow B.$$

Their widths can nevertheless differ arbitrarily. For example, the undirected $n \times n$ grid with $n \geq 2$ has a single edge as its core, even though its fractional hypertree width grows with n .

Observe that in the $\text{Hom}(\mathcal{H})$ setting this has no effect, as every hypergraph H admits a structure A with $H_A = H$ where A is a core. This motivates an alternative version of the problem where we restrict the patterns to a class $\mathcal{A}$ of finite relational structures.

Hom($\mathcal{A}$)

Input Finite relational structures A and B over the same signature, with $A \in \mathcal{A}$.

Question Does $A \rightarrow B$?

We write $p\text{-Hom}(\mathcal{A})$ for the parameterisation of $\text{Hom}(\mathcal{A})$ by the pattern A and write $\text{core}(\mathcal{A})$ for the class of cores of structures in $\mathcal{A}$. Grohe [14] showed that, for any recursively enumerable class $\mathcal{A}$ of structures of bounded arity, bounded treewidth of $\text{core}(\mathcal{A})$ characterises both polynomial-time solvability and fixed-parameter tractability, assuming $\text{FPT} \neq \text{W}[1]$. We extend this characterisation to bounded incidence degeneracy using fractional hypertree width of the cores.

The main challenge over the hypergraph setting is the upper bound. Recall that the efficient algorithm for deciding $A \rightarrow B$ with exponent $O(k)$ requires an explicit width- k decomposition for

A . Even in the $\text{Hom}(\mathcal{H})$ setting, obtaining such a decomposition in polynomial time requires an approximation algorithm. In the $\text{Hom}(\mathcal{A})$ setting, this is not feasible as finding the core itself is already NP-hard [6].

To avoid this issue, Dalmau, Kolaitis, and Vardi [9] showed how to use existential pebble games to decide $A \rightarrow B$ in polynomial time when the cores have bounded treewidth, without having to compute the cores. Chen and Dalmau [7] extend this approach to bounded ghw of the cores. To apply it to fractional hypertree width, we need to bound the number of edges needed to cover each bag, rather than just their total fractional weight. Chen et al. [8, Section 4] use bounds on the integrality gap of edge covers to show that, under bounded VC dimension, bounded fhw implies bounded ghw. This transfers the upper bound without constructing a decomposition of the core. Bounded incidence degeneracy ensures bounded VC dimension, giving the following proposition.

Proposition 4.2. *Let $\mathcal{A}$ be a class of finite relational structures of bounded incidence degeneracy. If $\text{core}(\mathcal{A})$ has bounded fractional hypertree width, then $\text{Hom}(\mathcal{A}) \in \text{PTIME}$.*

Proof. By [8, Lemma 2 and Section 4, Corollary 1], bounded VC dimension of $\{H_A : A \in \mathcal{A}\}$ and bounded fractional hypertree width of $\text{core}(\mathcal{A})$ imply $\text{Hom}(\mathcal{A}) \in \text{PTIME}$. Let an integer d bound the incidence degeneracy. The incidence graphs contain no $K_{d+1, d+1}$, so $\text{VCdim}(H_A) \leq d + \lceil \log_2(d+1) \rceil$ for every $A \in \mathcal{A}$ by [4, Table 1 and Appendix B]. $\square$

Importantly, the proposition requires bounded incidence degeneracy, and we do not know whether it holds for bounded λ in general. This remains an interesting gap between the following Theorem 4.3 and Theorem 4.1.

Theorem 4.3. *Let $\mathcal{A}$ be a recursively enumerable class of finite relational structures with bounded incidence degeneracy. Assuming ETH, the following are equivalent.*

- (i) $\text{core}(\mathcal{A})$ has bounded fractional hypertree width.
- (ii) $\text{core}(\mathcal{A})$ has bounded submodular width.
- (iii) $p\text{-Hom}(\mathcal{A}) \in \text{FPT}$.
- (iv) $\text{Hom}(\mathcal{A}) \in \text{PTIME}$.

Proof. For $A \in \mathcal{A}$, set $C = \text{core}(A)$ and let d bound the incidence degeneracy of $\mathcal{A}$. We may choose C as a substructure of A . Thus $I(H_C)$ is a subgraph of $I(H_A)$, and $\lambda(H_C) \leq \mu(H_C) \leq d$. For $\text{subw}(C) \geq 2$, the width hierarchy and Theorem 1.1 give

$$\text{subw}(C) \leq \text{fhw}(C) \leq O_d(\text{subw}(C) \log \text{subw}(C)).$$

For $\text{subw}(C) < 2$, the same theorem gives $\text{fhw}(C) = O_d(1)$. This proves (i) $\Leftrightarrow$ (ii). By Chen et al. [8, Theorem 1 and Lemma 2], $p\text{-Hom}(\mathcal{A}) \in \text{FPT}$ if and only if $\text{core}(\mathcal{A})$ has bounded submodular width under ETH, giving (ii) $\Leftrightarrow$ (iii). Proposition 4.2 gives (i) $\Rightarrow$ (iv). Every polynomial-time algorithm is fixed-parameter tractable, giving (iv) $\Rightarrow$ (iii). $\square$

Bounded incidence degeneracy includes both bounded rank and bounded vertex degree, and also allows both to be unbounded when the incidence graphs are sparse. The following corollary gives several standard settings in which the characterisation applies.

Corollary 4.4. *Let $\mathcal{A}$ be a recursively enumerable class of finite relational structures. Each of the following conditions implies the equivalences in Theorem 4.3, assuming ETH.*

- (i) The hypergraphs H_A have bounded rank.
- (ii) The hypergraphs H_A have bounded maximum vertex degree.

- (iii) The hypergraphs H_A are Zykov-planar, i.e., the incidence graphs $I(H_A)$ are planar (see [37]).
- (iv) The incidence graphs $I(H_A)$ exclude a fixed graph as a topological minor.

Proof. For every hypergraph H , its incidence degeneracy is at most its rank and at most its maximum vertex degree. Thus (i) and (ii) bound incidence degeneracy. In (iii), every incidence subgraph is planar and bipartite. Euler’s formula gives average degree less than 4 in every nonempty such subgraph, so the incidence graphs are 3-degenerate. For (iv), let F be the excluded topological minor and set $t = |V(F)|$. Bollobás and Thomason [2] proved that, for a universal constant c , every graph of average degree at least ct^2 contains a subdivision of K_t , and hence of F . Every nonempty subgraph of an incidence graph therefore has average degree less than ct^2 , giving bounded incidence degeneracy. $\square$

5 Implications for Counting and the Polymatroid Width Hierarchy

We apply the main width theorem to open questions in exact counting and the polymatroid width hierarchy. The comparison bounds the gap between the widths used by PANDA and #PANDA by a multiplicative $O(\lambda \log \max\{2, \text{subw}\})$ factor. On bounded- λ classes, it follows that fixed-parameter tractability of decision implies polynomial-time exact counting. We then show that the intermediate polymatroid widths lie within the same multiplicative band.

The counting question asks whether fixed-parameter tractability of deciding $A \rightarrow B$ implies fixed-parameter tractability of counting all homomorphisms from A to B . For hypergraph restrictions, Marx’s theorem reduces this question to whether bounded submodular width suffices for exact counting. This remains open in general.

Throughout this section, counting means counting all homomorphisms, or equivalently all answers to a full conjunctive query. Write $\#\text{Hom}(\mathcal{H})$ for this counting problem on patterns whose hypergraphs lie in $\mathcal{H}$, and $p\text{-}\#\text{Hom}(\mathcal{H})$ for its parameterisation by the pattern.

The algorithmic gap appears in PANDA, which evaluates Boolean queries in time $\tilde{O}(N^{\text{subw}(H)})$ for a fixed query hypergraph H and database size N [22]. PANDA uses different decompositions for different parts of the data. The resulting sets of satisfying assignments can overlap, which is harmless for decision but prevents us from simply summing their counts. Whether exact counting can achieve the same exponent remains open [24, Section 8]. The weaker question of whether bounded subw suffices for fixed-parameter counting also remains open in general [17, 19, 24].

Abo Khamis et al. [19] introduce #PANDA, which ensures that each satisfying assignment contributes through exactly one decomposition. Its exponent is sharp-submodular width $\#\text{subw}$. This condition is obtained by restricting the submodularity inequalities available to the algorithm. Let $\mathcal{S}_H^\#$ consist of the nonnegative, monotone, edge-dominated functions b with $b(\emptyset) = 0$ satisfying

$$b(X \cup Y) + b(X \cap Y) \leq b(X) + b(Y) \quad \text{whenever } X \cap Y \subseteq e \text{ for some } e \in E(H).$$

Then $\#\text{subw}(H) = \text{width}_{\mathcal{S}_H^\#}(H)$, and #PANDA evaluates scalar aggregates over a commutative semiring in $\tilde{O}(N^{\#\text{subw}(H)})$ semiring operations for fixed H [19].

Requiring submodularity only for intersections contained in an edge *enlarges* the family of functions over which width is maximised. Thus the stronger algorithmic requirement for counting is reflected in a potentially larger width. The known inequalities are [19, Proposition 3.15]

$$\text{subw}(H) \leq \#\text{subw}(H) \leq \text{fhw}(H).$$

These inequalities alone do not bound $\#\text{subw}$ in terms of subw , since bounded submodular width can coexist with unbounded fractional hypertree width [29, 30]. Our comparison bounds the gap once $\lambda(H)$ is taken into account.

Corollary 5.1. *For every hypergraph H with $s = \text{subw}(H)$,*

$$s \leq \#\text{subw}(H) \leq \text{fhw}(H) \leq C\lambda(H)s \log \max\{2, s\}$$

for a universal constant C .

Thus, for $\text{adw}(H) \geq 2$, the ratio between the widths governing $\#\text{PANDA}$ and PANDA satisfies

$$\frac{\#\text{subw}(H)}{\text{subw}(H)} \leq C\lambda(H) \log \text{adw}(H).$$

This bounds the exponent needed for exact counting by an $O(\lambda(H) \log \text{adw}(H))$ factor relative to PANDA 's decision exponent. On bounded- λ classes, it also settles the corresponding tractability question. Bounded submodular width implies polynomial-time exact counting.

Corollary 5.2. *Let $\mathcal{H}$ be a recursively enumerable hypergraph class of bounded λ . Assuming ETH, the following are equivalent.*

- (i) $\mathcal{H}$ has bounded submodular width.
- (ii) $p\text{-Hom}(\mathcal{H}) \in \text{FPT}$.
- (iii) $p\text{-}\#\text{Hom}(\mathcal{H}) \in \text{FPT}$.
- (iv) $\#\text{Hom}(\mathcal{H}) \in \text{PTIME}$.

Proof. As in [Theorem 4.1](#), (i) $\Leftrightarrow$ (ii) under ETH. For (i) $\Rightarrow$ (iv), [Theorem 1.1](#) gives bounded fractional hypertree width. A decomposition of bounded width can be found in polynomial time [\[28\]](#), and dynamic programming on it counts all homomorphisms in polynomial time [\[10, 32\]](#). The implication (iv) $\Rightarrow$ (iii) is immediate, and (iii) $\Rightarrow$ (ii) follows by testing whether the count is positive. $\square$

On bounded- λ classes, the implications (i) $\Rightarrow$ (iv) $\Rightarrow$ (iii) $\Rightarrow$ (ii) hold unconditionally. ETH is used only for (ii) $\Rightarrow$ (i). The corollary applies to all the structural classes in [Corollary 4.4](#).

The same bound also controls the intermediate polymatroid widths. Let $\mathcal{N}_H$ be the edge-dominated normal polymatroids, namely nonnegative combinations of the functions $h_S(X) = \mathbf{1}_{X \cap S \neq \emptyset}$ for $\emptyset \neq S \subseteq V(H)$ [\[20\]](#). Let $\mathcal{E}_H$ be the edge-dominated entropy functions of finite discrete random variables, and let $\bar{\mathcal{E}}_H$ consist of the edge-dominated limits of entropy functions. We have

$$\mathcal{M}_H \subseteq \mathcal{N}_H \subseteq \mathcal{E}_H \subseteq \bar{\mathcal{E}}_H \subseteq \mathcal{S}_H \subseteq \mathcal{S}_H^\#.$$

Each of these function families defines a width parameter through $\text{width}_{\mathcal{F}}(H)$, just as modular and submodular functions define adaptive and submodular width. The intermediate cones are not merely formal interpolations. Abo Khamis et al. [\[21\]](#) use normal, almost-entropic, and polymatroid optimisation to obtain join-size bounds from ℓ_p -norms of degree sequences. These choices expose a trade-off between realisable worst-case instances, asymptotic tightness, and efficient LP computation. For simple degree sequences all three bounds coincide, with normal relations witnessing tightness. Almost-entropic optimisation also gives the entropic width introduced with PANDA [\[18\]](#), whose attainment as a query-evaluation exponent remains open [\[17\]](#). Normal polymatroids also underlie decidable cases of bag query containment [\[20\]](#) and the computation of polymatroid bounds under simple degree constraints [\[16\]](#). Our comparison bounds how far the decomposition widths obtained from these function families can lie apart.

Corollary 5.3. *For every hypergraph H with $a = \text{adw}(H) \geq 2$,*

$$\begin{aligned} a &\leq \text{width}_{\mathcal{N}_H}(H) \leq \text{width}_{\mathcal{E}_H}(H) \leq \text{width}_{\bar{\mathcal{E}}_H}(H) \\ &\leq \text{subw}(H) \leq \#\text{subw}(H) \leq \text{fhw}(H) \leq C\lambda(H) a \log a. \end{aligned} \tag{8}$$

The proof is once again immediate from the inclusions and [Theorem 1.1](#). Since every term is at least a , the entire hierarchy lies within an $O(\lambda(H) \log a)$ multiplicative band. Marx’s bound $\text{subw}(H) = O(a^4)$ [30] already links the boundedness of the widths through subw without dependence on the hypergraph. This is already considered surprising, as modular functions can somehow always certify large width attained by submodular functions. Indeed, so far no clear explanation for this phenomenon (beyond the technical argument that yielded the bound) is known. In the bounded- λ setting, our results show that the gap is much smaller than quartic. In general, the bound provides additional insight into which structural factors play a role in this connection between modular and submodular functions.

6 Related Work

Previous work on adw/fhw comparisons Lanzinger [25] characterises tractability for recursively enumerable classes of hypergraphs of maximum degree two by bounded generalised hypertree width, assuming $\text{FPT} \neq \text{W}[1]$. Together with Marx’s characterisation [30], this implies, under ETH, that bounded fractional hypertree width and bounded submodular width coincide on such classes. This comparison is qualitative and follows through the complexity characterisations.

In recent work, Lanzinger [26] uses multicommodity flows to obtain, for $\text{adw}(H) \geq 2$,

$$\text{ghw}(H) = O(\text{adw}(H) \log \text{adw}(H))$$

on classes of bounded edge excess and on hypergraphs with private intersections. The constants may depend on the edge-excess bound. Although the main statements use submodular width, the modular witnesses in [26, Section 6.1] give the same lower bound for adaptive width. We write the equivalent upper bound on ghw to make the comparison with [Theorem 1.1](#) explicit. Bounded edge excess here means that, in each hyperedge, the sum of the amounts by which vertex degrees exceed two is bounded by a fixed constant. A hypergraph has private intersections if every nonempty intersection of two distinct hyperedges contains a vertex belonging to exactly those two hyperedges.

[Theorem 1.1](#) gives an unconditional quantitative comparison for every bounded-degree class and, more generally, every class of bounded incidence degeneracy. For $\text{adw}(H) \geq 2$,

$$\text{fhw}(H) = O(\Delta(H) \text{adw}(H) \log \text{adw}(H)).$$

Thus our result extends the degree-two and bounded-edge-excess comparisons to every bounded-degree class. The two bounds control different widths, but

$$\text{fhw}(H) \leq \text{ghw}(H) \leq \Delta(H) \text{fhw}(H).$$

Indeed, in a fractional cover, each covered vertex belongs to an edge of weight at least $1/\Delta(H)$. Choosing all such edges gives an integral cover of at most $\Delta(H)$ times the fractional cost. Bounded edge excess implies bounded degree, whereas bounded degree allows arbitrarily many degree-three vertices in one hyperedge.

The private intersection case remains complementary, since that condition permits unbounded incidence degeneracy and unbounded λ . Its degree-two witnesses need not survive vertex restriction, whereas our separator recursion requires rounding guarantees on every induced subhypergraph. The flow and separator arguments therefore apply under different structural conditions. We discuss this obstruction in [Section 7.1](#).

Connections to Korchemna et al. [23] Finally, [Theorem 1.1](#) relies on important results of Korchemna et al. [23]. They derive these results while developing a novel polynomial-time approximation algorithm for fractional hypertree width. For a width bound $k \geq 2$, their guarantee includes a decomposition of width $O(\lambda(H)k \log k)$ whenever $\text{fhw}(H) \leq k$. Their proof introduces the fractional balanced-separator relaxation and the rounding theorem used in [Section 3](#). Korchemna et al. combine this rounding theorem with a recursive separator construction to approximate fhw , using $\text{fhw}(H)$ itself to bound the optimum separator cost. Our minimax argument instead bounds that cost by $\text{adw}(H)$ uniformly over all induced subhypergraphs and hyperedge weightings, which turns their rounding theorem into a direct comparison between the two widths. In particular, the incidence degeneracy and primal independence terms in $\lambda(H)$ come from their rounding analysis. We prove $\text{fsep}_{1/2}(H) \leq \text{adw}(H)$ by minimax and the balanced-bag property of tree decompositions.

7 Conclusion

We proved that every hypergraph H with $\text{adw}(H) \geq 2$ satisfies

$$\text{fhw}(H) = O(\lambda(H)\text{adw}(H) \log \text{adw}(H)).$$

Together with the known width and complexity characterisations, this shows that on recursively enumerable hypergraph classes of bounded λ , bounded adaptive width, bounded submodular width, bounded fractional hypertree width, fixed-parameter tractability of $p\text{-Hom}$, and polynomial-time solvability of Hom are equivalent under ETH. The same characterisation holds for recursively enumerable classes of structures with bounded incidence degeneracy after taking cores. The width bound also gives polynomial-time exact counting on bounded- λ classes of bounded submodular width. Without a bound on λ , it gives a quasipolynomial-time algorithm for Hom whenever $p\text{-Hom}$ is fixed-parameter tractable, assuming ETH.

7.1 Open Problems

Unifying the flow and separator proofs. Our separator argument and the multicommodity-flow argument of [\[26\]](#) give closely related comparisons under incomparable structural hypotheses. For $\text{adw}(H) \geq 2$, they yield

$$\text{fhw}(H) = O(\lambda(H)\text{adw}(H) \log \text{adw}(H))$$

and, for hypergraphs with private intersections

$$\text{ghw}(H) = O(\text{adw}(H) \log \text{adw}(H)).$$

The private intersections condition permits unbounded λ and is not preserved by vertex restriction. Indeed, every hypergraph is an induced subhypergraph of one with private intersections, obtained by adding, for each pair of intersecting edges, a new vertex belonging to exactly those two edges. Thus a rounding guarantee depending only on fractional separator cost and valid on every induced subhypergraph cannot account for the private-intersection result. A common proof would instead have to use the private witnesses globally, or replace recursive restriction by a decomposition principle compatible with both vertex separators and edge cuts.

The logarithmic losses also enter at different points. In our proof they come entirely from rounding a fractional balanced separator of cost r to an integral separator of cost $O(\lambda(H)r \log r)$. The subsequent separator recursion loses only a constant factor. In the flow argument the logarithm

comes from the $O(\log m)$ node-capacitated flow–cut gap for product demands on a well-linked set of size m . It is unlikely that either approach can be strengthened to avoid the additional logarithmic factor.

No example is currently known in which either logarithmic loss is necessary. This leaves open the possibility that $\text{fhw}(H) = O(\lambda(H)\text{adw}(H))$ holds for all hypergraphs and that $\text{ghw}(H) = O(\text{adw}(H))$ holds for hypergraphs with private intersections. Proving either bound appears to require a more direct structural argument. Conversely, a matching logarithmic separation would identify a genuine obstruction shared by the two settings.

Further classes admitting separator rounding. The inequality $\text{fsep}_{1/2}(H) \leq \text{adw}(H)$ holds for every hypergraph. A broader problem is to identify structural conditions beyond bounded λ that permit effective rounding of these fractional separators. If a class admits a nondecreasing function g and a fixed $\eta \in [1/2, 1)$ such that every fractional $(\gamma, 1/2)$ -balanced separator x in every induced subhypergraph J can be rounded to a (γ, η) -balanced separator S with $\rho_J^*(S) \leq g(\rho_J^*(x))$, then the separator recursion from [Section 3.3](#) gives

$$\text{fhw}(H) \leq \frac{2(2 - \eta)}{1 - \eta} g(\text{adw}(H)).$$

Thus further rounding theorems would yield new width comparisons and new classes on which FPT decision implies polynomial-time exact counting, by the argument of [Corollary 5.2](#).

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